Problem · Unit 2 · 10 points · 60 min · High difficulty / trap

P09 — Bead on a vertical hoop held by a spring: reaction, equilibria and a pitchfork bifurcation

A smooth circular hoop of radius a and centre C is fixed in a vertical plane. A bead P of mass m is threaded on it (it can neither leave the hoop nor the plane). A linear spring of stiffness k and natural length l_0 joins P to the fixed point A, which lies on the vertical through C at the height $d = 2a$ above C. The natural length is $l_0 = d - a = a$, so the spring is unstretched when the bead is at the top T of the hoop. Gravity $-g\,\mathbf{j}$ acts. The position of the bead is given by the angle θ between $\mathbf{CT}$ (upward vertical) and $\mathbf{CP}$, positive clockwise in the figure (towards $+x$).

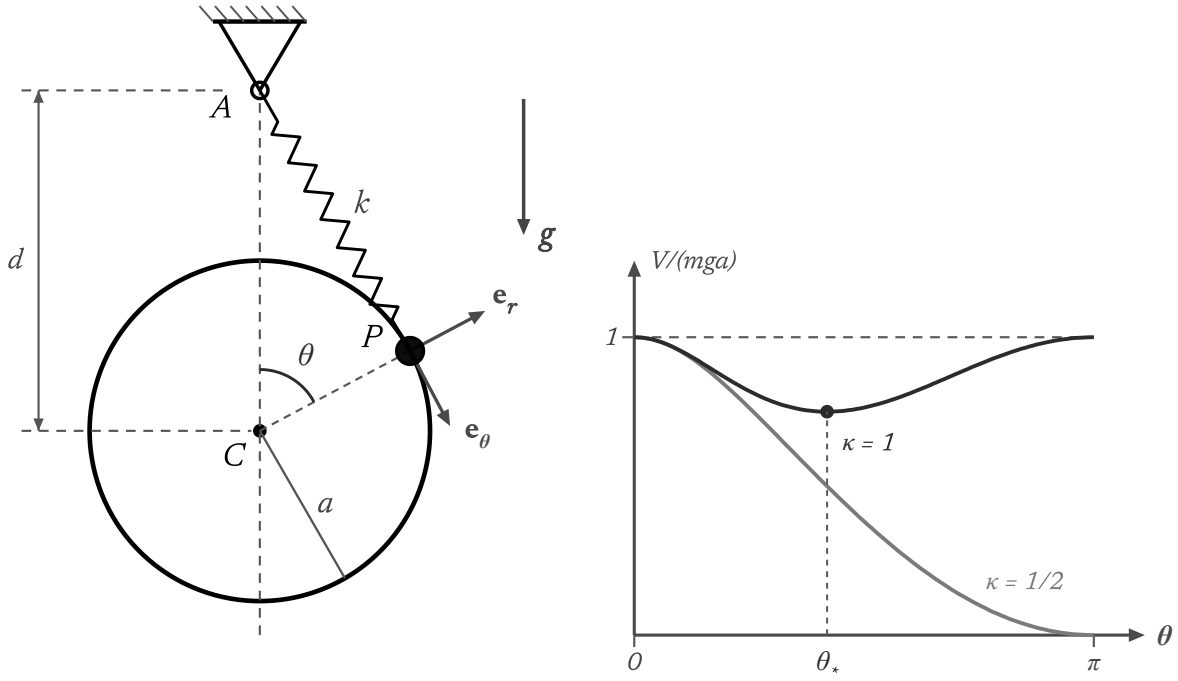


Figure 1. Bead P on the smooth hoop (centre C, radius a), spring of stiffness k from the fixed point A at height $d = 2a$ above C, angle θ from the top; right: $V(\theta)/(mga)$ for $\kappa = ka/(mg) = 1/2$ and $\kappa = 1$.

Use the inertial frame $S_0 = \{C; \mathbf{i}, \mathbf{j}\}$ ($\mathbf{i}$ horizontal, $\mathbf{j}$ upward) and the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$ (along $\mathbf{CP}$) and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}$ (tangent to the hoop, towards increasing θ). The results are expressed with the dimensionless stiffness

$$\kappa = \frac{ka}{mg}$$

Setup

- Data: m, a, k, g; $d = 2a$, $l_0 = a$. Numerical values in (e) and (f): $a = 0.40$ m, $g = 9.81$ m/s^2.
- Hypotheses: point particle; smooth, rigid, fixed hoop (bilateral constraint); massless linear spring that can be stretched or compressed; S_0 inertial.
- Spring length: $l = |\mathbf{AP}|$ with $\mathbf{AP} = a\,\mathbf{e}_r - d\,\mathbf{j}$.

Part (a) · 1.5 points

Degrees of freedom, geometry and potential energy

How many degrees of freedom has the bead? Find the spring length $l(\theta)$ and the total potential energy $V(\theta)$; check that the applied forces are conservative.

Step 1. Degrees of freedom

A free particle has 3 coordinates. The hoop is a curve: two holonomic, scleronomic constraints (the plane and the circle $|\mathbf{CP}| = a$), so 1 degree of freedom remains, described by θ (Notes §2.7).

Step 2. Spring length

$$l^2 = \mathbf{AP}\cdot\mathbf{AP} = (a\,\mathbf{e}_r - d\,\mathbf{j})\cdot(a\,\mathbf{e}_r - d\,\mathbf{j}) = a^2 - 2ad\,(\mathbf{e}_r\cdot\mathbf{j}) + d^2 = a^2 + d^2 - 2ad\cos\theta$$

With $d = 2a$: $l = a\sqrt{5 - 4\cos\theta}$, from $l(0) = a$ at the top to $l(\pi) = 3a$ at the bottom.

Step 3. Potentials

Weight: uniform, $V_g = mg\,y_P = mga\cos\theta$. Spring: $V_s = \tfrac{1}{2}k(l - l_0)^2$, a function of the position only (Notes §4.4). Both are conservative, the reaction of the smooth fixed hoop does no work (part (b)), so

$$V(\theta) = mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2, \qquad \frac{V}{mga} = \cos\theta + \frac{\kappa}{2}\Big(\sqrt{5 - 4\cos\theta} - 1\Big)^2$$

KEY RESULTS

- 1 degree of freedom, θ (the hoop removes 2 of the 3 spatial coordinates).
- $l^2 = a^2 + d^2 - 2ad\cos\theta = a^2(5 - 4\cos\theta)$; $l(0) = a = l_0$, $l(\pi) = 3a$.
- $V(\theta) = mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2$, i.e. $\dfrac{V}{mga} = \cos\theta + \dfrac{\kappa}{2}\Big(\sqrt{5 - 4\cos\theta} - 1\Big)^2$.

Part (b) · 2 points

Equation of motion and reaction of the hoop

Write the kinematics of P in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, the forces, the equation of motion free of the reaction, and the reaction $\mathbf{N}$ as a function of θ and $\dot\theta$. Check the equation of motion against $V(\theta)$.

Step 1. Change of basis

The pair $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ turns with θ. Since θ grows clockwise (from $\mathbf{j}$ towards $\mathbf{i}$), the right-handed order in the plane is $\{\mathbf{e}_\theta, \mathbf{e}_r\}$: with $\mathbf{k} = \mathbf{i} \times \mathbf{j}$, $\mathbf{e}_\theta \times \mathbf{e}_r = (\cos\theta\cos\theta + \sin\theta\sin\theta)\,\mathbf{k} = \mathbf{k}$ (the order $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ has determinant -1). Then

$$\begin{pmatrix}\mathbf{e}_\theta \\ \mathbf{e}_r\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i} \\ \mathbf{j}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta\end{pmatrix} = I$ and $\det[R] = \cos^2\theta + \sin^2\theta = +1$.

Step 2. Kinematics

$\mathbf{r}_{CP} = a\,\mathbf{e}_r$. By the chain rule, $\dot{\mathbf{e}}_r = \dot\theta(\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = \dot\theta(-\sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}) = -\dot\theta\,\mathbf{e}_r$. With a constant:

$$\mathbf{v}_P = a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_P = a\ddot\theta\,\mathbf{e}_\theta + a\dot\theta\,\dot{\mathbf{e}}_\theta = a\ddot\theta\,\mathbf{e}_\theta - a\dot\theta^2\,\mathbf{e}_r$$

Step 3. Forces

- Weight: $\mathbf{W} = -mg\,\mathbf{j} = -mg\cos\theta\,\mathbf{e}_r + mg\sin\theta\,\mathbf{e}_\theta$ (using $\mathbf{j}\cdot\mathbf{e}_r = \cos\theta, \mathbf{j}\cdot\mathbf{e}_\theta = -\sin\theta$).
- Spring: $\mathbf{F}_s = -k\,\dfrac{l - l_0}{l}\,\mathbf{AP} = -k\Big(1 - \dfrac{l_0}{l}\Big)\big[(a - d\cos\theta)\,\mathbf{e}_r + d\sin\theta\,\mathbf{e}_\theta\big]$, since $\mathbf{AP}\cdot\mathbf{e}_r = a - d\cos\theta$ and $\mathbf{AP}\cdot\mathbf{e}_\theta = -d\,(\mathbf{j}\cdot\mathbf{e}_\theta) = d\sin\theta$.
- Reaction: the hoop is smooth, so $\mathbf{N}$ is orthogonal to the tangent $\mathbf{e}_\theta$ (Notes §6.5); its component normal to the plane balances nothing (all other forces are in the plane), so $\mathbf{N} = N\,\mathbf{e}_r$.

Step 4. Equation of motion (tangential projection)

$$ma\ddot\theta = mg\sin\theta - k\Big(1 - \frac{l_0}{l}\Big)d\sin\theta = \sin\theta\Big[mg - kd\Big(1 - \frac{l_0}{l}\Big)\Big]$$

N does not appear: this is the reaction-free equation.

Step 5. Reaction (radial projection)

$$-ma\dot\theta^2 = -mg\cos\theta - k\Big(1-\frac{l_0}{l}\Big)(a-d\cos\theta) + N$$

$$N = mg\cos\theta + k\Big(1-\frac{l_0}{l}\Big)(a-d\cos\theta) - ma\dot\theta^2$$

Step 6. Check with V

By the chain rule, $\dfrac{dV}{d\theta} = -mga\sin\theta + k(l-l_0)\dfrac{dl}{d\theta}$ and, differentiating $l^2 = a^2+d^2-2ad\cos\theta$ implicitly, $2l\,l' = 2ad\sin\theta$, so $l' = ad\sin\theta/l$. Hence

$$V'(\theta) = \sin\theta\Big[-mga + kad\Big(1-\frac{l_0}{l}\Big)\Big] = -a\cdot ma\ddot\theta$$

which is the energy form $ma^2\ddot\theta = -V'(\theta)$ of Step 4 (the Lagrange equation, an extension beyond the course notes, gives the same).

KEY RESULTS

- $\mathbf{a}_P = a\ddot\theta\,\mathbf{e}_\theta - a\dot\theta^2\,\mathbf{e}_r$; $\mathbf{N} = N\,\mathbf{e}_r$ (smooth hoop, planar forces).
- $ma\ddot\theta = \sin\theta\Big[mg - kd\Big(1-\dfrac{l_0}{l}\Big)\Big]$, equal to $-V'(\theta)/a$.
- $N = mg\cos\theta + k\Big(1-\dfrac{l_0}{l}\Big)(a-d\cos\theta) - ma\dot\theta^2$ (positive: outwards).

Part (c) · 2 points

Equilibrium positions

Find all the equilibrium positions as functions of κ. Show that besides $\theta = 0$ and $\theta = \pi$ there is a symmetric pair $\theta = \pm\theta^*$ only above a critical value κ_c, and give $\cos\theta^*(\kappa)$.

Step 1. Factorisation

From (b), with $d = 2a$, $l_0 = a$ and $k = \kappa mg/a$:

$$\frac{V'(\theta)}{mga} = \sin\theta\Big[-1 + 2\kappa\Big(1 - \frac{a}{l}\Big)\Big]$$

Equilibria are the zeros of V' (Notes §4.6).

Step 2. $\sin\theta = 0$

$\theta = 0$ and $\theta = \pi$ are equilibria for every κ: by symmetry, the spring force and the weight are both along the vertical there, so they have no tangential component.

Step 3. The symmetric pair

$-1 + 2\kappa(1 - a/l) = 0 \Rightarrow 1 - a/l = 1/(2\kappa) \Rightarrow l^* = \dfrac{a}{1 - 1/(2\kappa)} = \dfrac{2\kappa a}{2\kappa - 1}$, which requires $\kappa > \frac{1}{2}$ for $l^* > 0$. Substituting into $l^2 = a^2(5 - 4\cos\theta)$:

$$\cos\theta^* = \frac{1}{4}\Big(5 - \frac{l^{*2}}{a^2}\Big) = \frac{1}{4}\Big(5 - \frac{4\kappa^2}{(2\kappa - 1)^2}\Big)$$

Step 4. Existence condition

A point of the hoop has $a \le l \le 3a$; a new equilibrium needs $a < l^* < 3a$. $l^* > a$ holds for every $\kappa > \frac{1}{2}$. $l^* < 3a \iff 2\kappa < 3(2\kappa - 1) \iff \kappa > \frac{3}{4}$. So the pair $\pm\theta^*$ exists only for $\kappa > \kappa_c = \frac{3}{4}$; at $\kappa = \frac{3}{4}$, $\cos\theta^* = \frac{1}{4}(5 - 9) = -1$: the pair is born **at the bottom**. As $\kappa \to \infty$, $l^* \to a$ and $\theta^* \to 0$.

KEY RESULTS

- $\theta = 0$ (top) and $\theta = \pi$ (bottom) for every κ.
- $G = 0 \iff 1 - \dfrac{a}{l} = \dfrac{1}{2\kappa} \iff l^* = \dfrac{2\kappa a}{2\kappa - 1}$; $\cos\theta^* = \dfrac{1}{4}\Big(5 - \dfrac{4\kappa^2}{(2\kappa - 1)^2}\Big)$.
- The pair exists only for $\kappa > \kappa_c = \frac{3}{4}$ ($\theta^* \to \pi$ as $\kappa \to \frac{3}{4}^+$, $\theta^* \to 0$ as $\kappa \to \infty$).

Part (d) · 1.5 points

Stability and the bifurcation

Classify the stability of each equilibrium in the two regimes $\kappa < \frac{3}{4}$ and $\kappa > \frac{3}{4}$. Describe what happens at $\kappa = \frac{3}{4}$. Is "the bottom of the hoop is always stable" true?

Step 1. Second derivative

From $V' = \sin\theta\, G(\theta)$ with $G = -mga + kad(1 - l_0/l)$, the product rule gives $V'' = \cos\theta\, G + \sin\theta\, G'$, and by the chain rule $G' = kad\,\dfrac{l_0}{l^2}\,l' = kad\,\dfrac{l_0}{l^2}\,\dfrac{ad\sin\theta}{l}$.

Step 2. Top and bottom

At $\theta = 0$: $l = l_0$, so $G(0) = -mga$ and $V''(0) = -mga < 0$: unstable for every κ (the spring is slack there and cannot hold the bead up).

At $\theta = \pi$: $l = 3a$, $G(\pi) = -mga + k\,a\,2a\,\frac{2}{3} = -mga + \frac{4}{3}ka^2$ and $V''(\pi) = \cos\pi\, G(\pi) = mga - \frac{4}{3}ka^2 = mga\left(1 - \frac{4}{3}\kappa\right)$.

Step 3. The pair

At θ^*, $G = 0$, so $V''(\theta^*) = \sin\theta^* G'(\theta^*) = \dfrac{ka^2d^2l_0\sin^2\theta^*}{l^{*3}} > 0$: a minimum, stable.

Step 4. The two regimes (the trap)

- $\kappa < \frac{3}{4}$ (soft spring): two equilibria; top unstable, **bottom stable**.
- $\kappa > \frac{3}{4}$ (stiff spring): four equilibria; top unstable, **bottom unstable**, $\pm\theta^*$ stable. The stretched spring pulls the bead up harder than gravity pulls it down, and the bead settles on one side.

The bottom is not always stable. At $\kappa = \frac{3}{4}$, $V''(\pi) = 0$ and $V'''(\pi) = 0$ (symmetry), and $V''''(\pi) = \frac{1}{3}mga > 0$: the bottom is still a (flat) minimum. As κ crosses $\frac{3}{4}$ the minimum at the bottom splits into two minima that move away symmetrically: a **supercritical pitchfork bifurcation** (see the energy curves of the figure).

KEY RESULTS

- Top: $V''(0) = -mga < 0$, always unstable.
- Bottom: $V''(\pi) = mga\left(1 - \frac{4}{3}\kappa\right)$: stable for $\kappa < \frac{3}{4}$, **unstable** for $\kappa > \frac{3}{4}$.
- Pair: $V''(\theta^*) = \dfrac{ka^2d^2l_0\sin^2\theta^*}{l^{*3}} > 0$: stable whenever it exists. Supercritical pitchfork at $\kappa = \frac{3}{4}$ ($V''''(\pi) = \frac{1}{3}mga > 0$ there).

Part (e) · 1.5 points

Small oscillations

Compute the angular frequency of small oscillations (i) about the bottom for $\kappa = \frac{1}{2}$ and (ii) about θ^* for $\kappa = 1$ (give θ^*). Numerical values for $a = 0.40$ m.

Step 1. Linearisation

With $\theta = \theta_e + \eta$, $V'(\theta) = V'(\theta_e) + V''(\theta_e)\eta + O(\eta^2)$ and $V'(\theta_e) = 0$, so $ma^2\ddot{\eta} + V''(\theta_e)\eta = 0$ and $\omega^2 = V''(\theta_e)/(ma^2)$ when $V''(\theta_e) > 0$.

Step 2. Bottom, $\kappa = \frac{1}{2}$

$V''(\pi) = mga(1 - \frac{4}{3}\cdot\frac{1}{2}) = \frac{1}{3}mga$, so $\omega^2 = \dfrac{g}{3a} = \dfrac{9.81}{1.2} = 8.175\ \text{s}^{-2}$ and $\omega = 2.859$ rad/s.

Step 3. Side equilibrium, $\kappa = 1$

$l^* = 2a/(2-1) = 2a$, $\cos\theta^* = \frac{1}{4}(5-4) = \frac{1}{4}$, $\theta^* = \arccos\frac{1}{4} = 75.52^\circ$, $\sin^2\theta^* = \frac{15}{16}$. With $k = mg/a$, $d = 2a$, $l_0 = a$:

$$V''(\theta^*) = \frac{k\,a^2\,(4a^2)\,a\cdot\frac{15}{16}}{8a^3} = \frac{15}{32}ka^2 = \frac{15}{32}mga, \qquad \omega^2 = \frac{15}{32}\frac{g}{a}$$

Numerically $\omega^2 = \frac{15}{32}\times\frac{9.81}{0.40} = 11.50\ \text{s}^{-2}$, $\omega = 3.391$ rad/s.

KEY RESULTS

- (i) $\omega^2 = \dfrac{g}{3a}$, $\omega = 2.859$ rad/s.
- (ii) $l^* = 2a$, $\cos\theta^* = \frac{1}{4}$, $\theta^* = 75.52^\circ$; $\omega^2 = \dfrac{15}{32}\dfrac{g}{a}$, $\omega = 3.391$ rad/s.

Part (f) · 1.5 points

Energy integral and the bead at the bottom

Write the energy first integral. For $\kappa = \frac{1}{2}$ the bead leaves the top with a vanishingly small speed. Find its speed at the bottom and the reaction of the hoop there. Would a bead released exactly at rest at the top move?

Step 1. First integral

The reaction is normal to $\mathbf{v}_P = a\dot\theta\,\mathbf{e}_\theta$, so its power $\mathbf{N}\cdot\mathbf{v}_P = 0$ (Notes §6.4 to §6.5); the other forces derive from V. Hence (Notes §4.5)

$$\tfrac{1}{2}ma^2\dot\theta^2 + mga\cos\theta + \tfrac{1}{2}k\big(l(\theta) - l_0\big)^2 = E$$

Check: $\dfrac{dE}{dt} = \dot\theta\big(ma^2\ddot\theta + V'(\theta)\big) = 0$ by (b).

Step 2. Top versus bottom

$V(0) = mga + 0$ and $V(\pi) = -mga + \frac{1}{2}k(3a-a)^2 = -mga + 2ka^2$, so $V(\pi) - V(0) = 2ka^2 - 2mga = 2mga(\kappa - 1)$.

Step 3. Speed at the bottom

$\theta = 0$ is an equilibrium: released **exactly** at rest there, the bead stays. With an infinitesimal push, $E = V(0)$. For $\kappa = \frac{1}{2} < \frac{3}{4}$ the only equilibria are 0 and π, so V decreases monotonically on $(0,\pi)$ and the bead reaches the bottom with

$$\tfrac{1}{2}mv^2 = V(0) - V(\pi) = 2mga(1-\kappa) \implies v^2 = 4ga(1-\kappa) = 2ga, \qquad v = \sqrt{2 \times 9.81 \times 0.40} = 2.801 \text{ m/s}$$

With no spring ($\kappa = 0$) it would be $v^2 = 4ga$: the spring stores part of the energy.

Step 4. Reaction at the bottom

At $\theta = \pi$: $l = 3a$, $1 - l_0/l = \tfrac{2}{3}$, $a - d\cos\pi = 3a$, $a\dot\theta^2 = v^2/a = 2g$ and $k = mg/(2a)$:

$$N = mg\cos\pi + \frac{mg}{2a}\cdot\frac{2}{3}\cdot 3a - m\cdot 2g = -mg + mg - 2mg = -2mg$$

$N < 0$ means the reaction points along $-\mathbf{e}_r$, towards C; at the bottom $-\mathbf{e}_r = \mathbf{j}$, so the hoop pushes the bead **up** with $2mg$. Check on the vertical: $N_{up} - mg + T_{spring} = mv^2/a$ with $T_{spring} = k(3a - a) = mg$ gives $N_{up} = 2mg - mg + mg = 2mg$.

KEY RESULTS

- $\tfrac{1}{2}ma^2\dot\theta^2 + mga\cos\theta + \tfrac{1}{2}k(l-l_0)^2 = E$; $V(\pi) - V(0) = 2mga(\kappa - 1)$.
- $\kappa = \tfrac{1}{2}$: $v^2_{bottom} = 4ga(1-\kappa) = 2ga$, $v = 2.801$ m/s (against $4ga$ without spring).
- $N(\pi) = -2mg$: the hoop pushes the bead **towards** C, i.e. upwards with $2mg$. At rest exactly at the top the bead stays (equilibrium); it needs a disturbance.

Interpretation and checks

- **Dimensional check:** $\kappa = ka/(mg)$ is (N/m)(m)/N = 1; V and mga in J; $\omega^2 = g/a$ in s^{-2}; N in N; $v^2 = ga$ in $\text{m}^2\ \text{s}^{-2}$.
- **Limits:** $\kappa \to 0$ recovers the bead on a hoop under gravity alone (top unstable, bottom stable, $\omega^2 = g/a$ at the bottom: our $V''(\pi)$ gives mga). $\kappa \to \infty$: the spring dominates and the stable positions approach the top, where the spring is unstretched. For $\kappa = 1$, $V(\pi) = V(0)$: a bead leaving the top with a vanishing speed approaches the (now unstable) bottom only asymptotically, never reaching it in finite time.
- **Common mistakes in the exam:** (1) assuming the bottom is always the stable position; (2) missing the side equilibria by cancelling $\sin\theta$ without keeping the second factor; (3) using $l - l_0$ with the wrong natural length (the spring is unstretched at the **top**, not at the bottom); (4) forgetting that the reaction of a bead on a hoop can point either way (bilateral constraint), and assuming $N \geq 0$; (5) "releasing from rest at the top" without noticing that it is an equilibrium.